Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"
Framework Service Contract N° 948120 – IPR – 2023

PRODUCT USER MANUAL

NORMALIZED DIFFERENCE VEGETATION INDEX
VERSION 3.0

Issue I1.02

Organization name of lead contractor for this deliverable: VITO

Book Captain:	Else Swinnen
Contributing Authors:	Carolien Toté
	Dominique De Munck

Dissemination Level		
PU	Public	X
PP	Restricted to other programme participants (including the Commission Services)	
RE	Restricted to a group specified by the consortium (including the Commission Services)	
CO	Confidential, only for members of the consortium (including the Commission Services)	

© C-GLOPS Lot1 consortium

Document Release Sheet

Book captain:	Else Swinnen (VITO)	Sign Else Swinnen	Date 19.10.2026
Approval:	Roselyne Lacaze (HYGEOS)	Sign Roselyne Lacaze	Date 20.10.2026
Endorsement:	Nadine Gobron (JRC)	Sign	Date
Distribution:	Public		

© C-GLOPS Lot1 consortium

Change Record

Date	Page(s)	Description of Change	Release
26.11.2025	All	Initial version	I1.00
12.10.2026	p.15, 20	Clarification on the land/water masking	I1.01
19.10.2026	p.24	Correction of QLAG explanation	I1.02

© C-GLOPS Lot1 consortium

TABLE OF CONTENTS

Executive Summary 11

1 Background of the document 12

1.1 Scope and Objectives 12

1.2 Content of the document 12

1.3 Related documents 13

1.3.1 Applicable documents 13

1.3.2 Input 13

1.3.3 External documents 14

2 Algorithm 15

2.1 Overview 15

2.2 Input Data 15

2.3 Retrieval Methodology 16

2.3.1 Background 16

2.3.2 Processing Steps 16

2.4 Limitations of the Product 19

2.5 Differences with Previous Versions 20

3 Product Description 22

3.1 File Naming 22

3.2 File Format 23

3.3 Product Content 23

3.3.1 Data File 23

3.3.2 Metadata 25

3.4 Product Characteristics 29

3.4.1 Projection and Grid Information 29

3.4.2 Spatial Information 29

3.4.3 Temporal Information 29

3.5 Data Policies 29

3.6 Access and Contacts 30

4 Quality assessment 31

5 Product Usage 33

5.1 Recommendations on using the quality flag 33

© C-GLOPS Lot1 consortium

5.2 Suitability for specific sectors and applications 33

6 References 35

7 Annex: Requirements 38

7.1 Specific technical details and requirements 38

7.2 Further requirements 38

7.2.1 Output product composition 38
7.2.2 Data Structure 39
7.2.3 Data format 39
7.2.4 Uncertainties and Validation 39
7.2.5 Input data 40
7.2.6 Product delivery 40

© C-GLOPS Lot1 consortium

List of Figures

Figure 1: Flow diagram of the NDVI 300m Version 3.0 algorithm. 17

© C-GLOPS Lot1 consortium

List of Tables

Table 1: Summary of key information 12
Table 2: Algorithm differences between successive versions of the NDVI 300m products 21
Table 3: Algorithm differences between successive versions of the NDVI 1km products 21
Table 4: Explanation in version numbering and recommendations for using efficiently the products 22
Table 5: Minimum, maximum values and associated resolution for NDVI product layers 23
Table 6: Description of the quality flags embedded into the NDVI layer 24
Table 7: Bit number, description and value of the quality flags. 25
Table 8: Description of file attributes for an OLCI-based NDVI file 26
Table 9: Description of layer attributes 28

© C-GLOPS Lot1 consortium

List of Acronyms

6S	Second Simulation of the Satellite Signal in the Solar Spectrum
AC	Atmospheric Correction
AD	Applicable Document
ASIS	Agricultural Stress Index System
ATBD	Algorithm Theoretical Basis Document
BRDF	Bidirectional Reflectance Distribution Function
C	Collection
CAL/VAL	Calibration/Validation
CEOS	Committee on Earth Observation System
CGLOPS	Copernicus Global Land Operations
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
DN	Digital Number
ECV	Essential Climate Variables
ENVISAT	Environmental Satellite
ESA	European Space Agency
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
FAO	Food and Agriculture Organisation of the United Nations
GBOV	Ground-Based Observations for Validation
GCOS	Global Climate Observing System
GEOGLAM	Group on Earth Observation Global Agriculture Monitoring
geoTIFF	Georeferenced Tagged Image File Format
GLAM	Global Agriculture Monitoring
INSPIRE	Innovation in Science Pursuit for Inspired Research
JRC	Joint Research Centre
LPV	Land Products Validation
MARS	Monitoring Agriculture Resources
MCD43A4	MODIS/Terra and Aqua Nadir BRDF-Adjusted Reflectance Daily L3
MODIS	Moderate Resolution Imaging Spectroradiometer
MVC	Maximum Value Composite
NDVI	Normalized Difference Vegetation Index

© C-GLOPS Lot1 consortium

NetCDF	Network Common Data Form
NIR	Near-infrared reflectance band
NOBS	Number of Observations
OLCI	Ocean and Land Colour Instrument
PROBA-V	VEGETATION sensor on board Project for On-Board Autonomy platform
PUM	Product User Manual
QAR	Quality Assessment Report
QFLAG	Quality Flag
ReBeLS	Regularised BRDF inversion for Land Surface
S2GM	Sentinel-2 Global Mosaics
S3	Sentinel-3
S10	10-daily synthesis
SLSTR	Sea and Land Surface Temperature Radiometer
SMAC	Simplified Method for Atmospheric Correction
SPOT	Système Pour l'Observation de la Terre
TOA	Top Of Atmosphere
TOC	Top Of Canopy
TOC-r	Top Of Canopy normalized reflectance
USDA	United State Department of Agriculture
VGT	VEGETATION sensor onboard SPOT4/5
VGT-S10	10-daily Top Of Canopy synthesis from VEGETATION sensor
VITO	Vlaamse Instelling voor Technologisch Onderzoek
WGS	World Geodetic System

© C-GLOPS Lot1 consortium

EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters take place in a timely manner and are complemented by the constitution of long-term time series.

The CLMS portfolio contains a global Normalized Difference Vegetation Index (NDVI) product at 300m spatial resolution. The NDVI Version 3.0 product is derived from the 10-daily Bidirectional Reflectance Distribution Function (BRDF)-normalized reflectances, generated from the data of the Ocean and Land Colour Instrument (OLCI) onboard the Sentinel-3 (S3) satellite and from Collection 2 (C2) of the PROBA-V (PV) data. The product consists of the NDVI layer, its associated uncertainty, the number of clear observations in the compositing window and a quality flag.

This document provides a detailed product description of the CLMS global NDVI version 3.0 (NDVI 300m V3.0), including a summary of the retrieval methodology, detailed information on file naming, file format, product content, product characteristics, data policies and data access, and a summary of the quality assessment.

© C-GLOPS Lot1 consortium

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

The Product User Manual (PUM) is the primary document that users have to read before handling the products. It gives an overview of the product characteristics, in terms of algorithm, technical characteristics, and main validation results. The key information is summarized in Table 1.

Table 1: Summary of key information

Input data The NDVI 300m V3.0 products are generated from PROBA-V C2 and Sentinel-3 OLCI Baseline Collection 004 (BC004) BRDF-corrected Top-Of-Canopy (TOC) reflectances.	See §2.2, p.15
Retrieval algorithm The processing consists of atmospheric correction, BRDF correction and NDVI computation.	See §2.3, p.16
File naming c_gls_<PROD>_<YYYYMMDDHHMM>_<AREA>_<SENSOR>_V<VERSION>	See §3.1, p.22
File format and product content a multi-band netCDF4 with NDVI, NDVI_unc, NOBS and QFLAG	See §3.2, p.23
Projection and grid information Regular lat/lon grid (plate carrée) with resolution 1/336°	See §3.4.1, p.29
Spatial information Longitude [180°W – 180°E], Latitude [85°N – 65°S]	See §3.4.2, p.29
Temporal information Three products per month, dated day 1, day 11 and day 21 of each month	See §3.4.3, p.29
Data policies Data access is free and open: https://land.copernicus.eu/en/data-policy	See §3.5, p.29
Data access More details on data access are available on https://land.copernicus.eu/en/getting-oriented-with-the-clms-website.	See §3.6, p.30
Contact CLMS helpdesk: https://land.copernicus.eu/en/contact-service-helpdesk	See §3.6, p.30

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 summarizes the retrieval methodology
- Chapter 3 describes the technical properties of the product
- Chapter 4 reports the main results of the quality assessment
- Chapter 5 gives insights about product usage

The requirements are recalled in Annex.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

1.3.2 Input

Document ID	Descriptor
CGLOPS1_ATBD_S3-AC-V1.1	Algorithm Theoretical Basis Document of the Atmospheric Correction applied on Sentinel-3 OLCI and SLSTR data
CGLOPS1_ATBD_BRDFCorrection300m-V1.6	Algorithm Theoretical Basis Document of the BRDF correction version 1.6
CGLOPS1_ATBD_NDVI300m-V3	Algorithm Theoretical Basis Document of the NDVI 300m Version 3.0 product
CGLOPS1_QAR_NDVI300m-V3	Report describing the results of the scientific quality assessment of the NDVI 300m Version 3.0 product
CGLOPS1_QAR_S3-CloudMask	Quality Assessment Report of Sentinel-3 OLCI and SLSTR cloud, cloud shadow and snow detections

1.3.3 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs requirements, World Meteorological Organization, Geneva, Switzerland (2022)

Available online at: https://library.wmo.int/idurl/4/58111

PROBA-V C2 ATBD AC	PROBA-V Collection 2 Algorithm Theoretical Baseline Document of the Atmospheric Correction.

Available online at: https://proba-v.vgt.vito.be/sites/probavvgt/files/downloads/PROBA-V_C2_Atmospheric_Correction_ATBD.pdf

PROBA-V C2 PUM	PROBA-V Collection 2 Products User Manual

Available online at: https://proba-v.vgt.vito.be/sites/probavvgt/files/downloads/PROBA-V_C2_Products_User_Manual.pdf

© C-GLOPS Lot1 consortium

2 ALGORITHM

2.1 OVERVIEW

NDVI, or Normalized Difference Vegetation Index, is a dimensionless index that is indicative for vegetation density and is calculated by comparing the visible and near-infrared sunlight reflected by the surface (reflectance). The NDVI has a long history of use in the remote sensing community for a wide range of applications (e.g., Baret and Guyot, 1991; Rouse et al., 1974; Tucker et al., 1981). NDVI is a proxy to quantify vegetation amount, is closely related to the Fraction of Absorbed Photosynthetically Active Radiation (FAPAR) and is little scale dependent.

The CLMS global NDVI 300m V3.0 product is based on data from the PROBA-V and the Sentinel-3A and -3B OLCI sensors.

2.2 INPUT DATA

The input data for the NDVI calculation are global 10-daily BRDF normalized surface reflectances (TOC-r) derived from the red and NIR band of PROBA-V and from 4 Sentinel-3/OLCI bands (Oa7, Oa8, Oa16 and Oa18) from TOC reflectance v2.3. More details on the PROBA-V data can be found in the PROBA-V C2 PUM for the general data description and use [PROBA-V C2 PUM], and the ATBD of atmospheric correction of PROBA-V C2 [PROBA-V C2 ATBD AC]. More details about the atmospheric correction applied on S3 data can be found in the ATBD [CGLOPS1_ATBD_S3-AC-V1.1]. The BRDF correction applied to reflectances of both sensors is described in the ATBD [CGLOPS1_ATBD_BRDFCorrection300m-V1.6].

The following layers are used from the BRDF-corrected TOC data for each of the spectral bands:

- the normalized surface reflectance (TOC-r),
- its associated uncertainty (TOC-r_unc)
- the number of clear observations in the dekad time window (Nobs)
- the number of clear observations in the dekad time window classified as snow ($Nobs_{snow}$)
- and the quality information layer (QUIL) which provides information on the quality of the BRDF-correction. The QUIL indicates the following:
  - whether the retrieval is based on observations in the dekad time window or not,
  - whether the inversion was successful,
  - the quality of the BRDF inversion based on an estimation of the goodness of fit,
  - whether the TOC-r is out of range (i.e. TOC-r < 0 or TOC-r >1).

TOC-r is only available for pixels classified as land. For PROBA-V products, distinction between land and water (open sea/ocean) is made based on the PROBA-V C2 input status map. For Sentinel-3/OLCI, the land/water classification is based on the OLCI IdePix land mask, which is a static layer delineating open sea/ocean, lakes, reservoirs and wide rivers.

2.3 RETRIEVAL METHODOLOGY

2.3.1 Background

Vegetation status is commonly monitored by using near-infrared (NIR) and red reflectances in different combinations (Tucker, 1979), the NDVI being the most widely used band combination. Long-term NDVI time series at global scale and with dekadal temporal cadence have been used to monitor the Earth vegetation dynamics. The assembling of such multi-annual NDVI time series is currently done by merging observations from sensors with very large swaths. Thus, the surface reflectance observations (used to compute NDVI) collected from the different sensors have been acquired under different viewing and illumination conditions due to the large swath of the sensors and their different overpass times.

First methods to remove the directional effects in multi-temporal NDVI series were empirical (e.g., Duggin et al., 1982). The Maximum Value Composite (MVC) (Holben, 1986) results in a temporal NDVI profile related to the biomass variations (Gutman, 1989), but implies that there is a risk to select NDVI values resulting from undetected cloudy observations in areas with persistent cloud cover (Eklundh, 1995; Moody and Strahler, 1994) or far off-nadir observations (Carreiras et al., 2003; Meyer et al., 1995; Moody and Strahler, 1994).

In order to remove directional effects from long-term NDVI time series, BRDF correction (i.e. approximating the surface reflectance anisotropy with a model and use it to estimate reflectances normalised to a common Sun-sensor geometry) is performed to the red and NIR reflectances prior to the NDVI computation.

2.3.2 Processing Steps

The processing of NDVI Version 3.0 comprises three main steps (see Figure 1). The atmospheric correction is first applied to the Sentinel-3 OLCI Top-Of-Atmosphere data. This step yields Top-of-Canopy reflectances for the different OLCI bands. As a second step, the BRDF correction is performed individually for each pixel of the 2 red and 2 NIR OLCI TOC reflectances. The BRDF model is used to estimate normalized TOC-r reflectances in a common Sun-sensor geometry (nadir viewing and Sun zenith angle at 10:00 a.m. local time), which are used, in a third step, to compute the NDVI. The subsequent steps are discussed in detail respectively in the following Algorithm Theoretical Design Documents: [CGLOPS1_ATBD_S3-AC-V1.1], [CGLOPS1_ATBD_BRDFCorrection300m-V1.6] and [CGLOPS1_ATBD_NDVI300m-V3]. The third step is briefly described below.

The same processing scheme is used for the archive PROBA-V data, except that the input data are already atmospherically corrected. The first processing step is the BRDF correction on the data [see CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. Since there is only 1 red and 1 NIR band, these bands are used to calculate the NDVI.

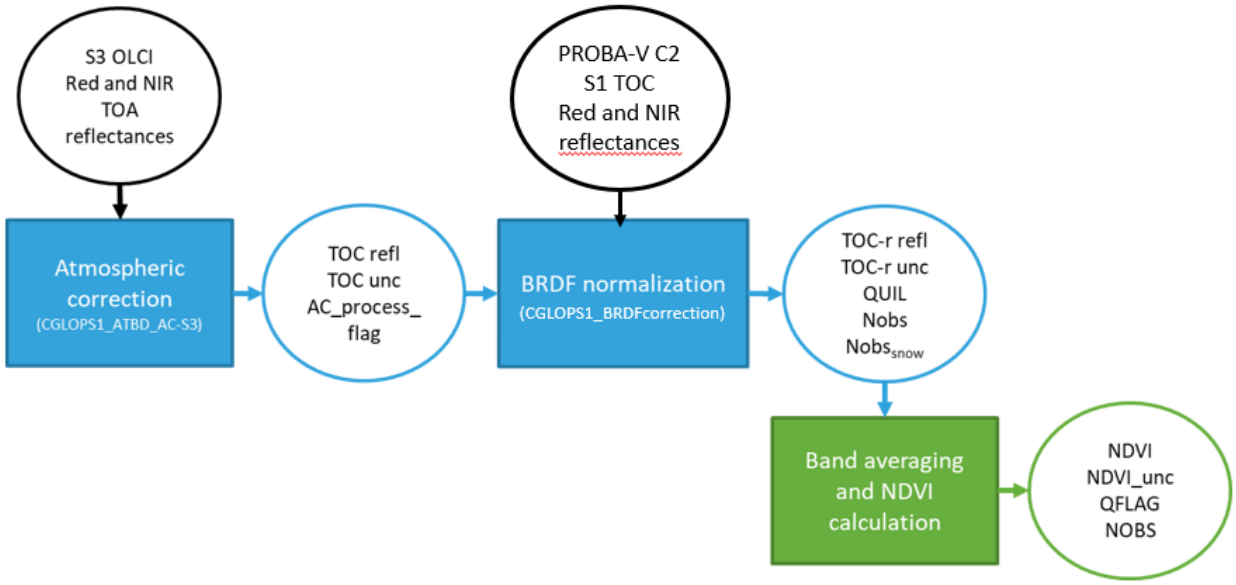


Figure 1: Flow diagram of the NDVI 300m Version 3.0 algorithm.

2.3.2.1 NDVI computation

Prior to NDVI calculation, 2 narrow red TOC-r bands and 2 narrow NIR TOC-r bands of S3 OLCI are converted to a broad red and NIR band by averaging using Eq 1 and Eq 2.

Calculate meanRED and meanNIR reflectances from the TOC-r.

$$meanRED = (B7 + B8)/2 \quad \textbf{Eq 1}$$

$$meanNIR = (B16 + B18)/2 \quad \textbf{Eq 2}$$

With *Bx* is the TOC-r reflectance of band *x*.

Then, the NDVI is calculated using the meanRED and meanNIR bands.

$$NDVI = \frac{meanNIR - meanRED}{meanNIR + meanRED} \quad \textbf{Eq 3}$$

For PROBA-V data, the NDVI is calculated in the same way based on the available red and NIR bands. The NDVI from PROBA-V is corrected by 4.5% to agree better the S3-based NDVI (see Annex in CGLOPS1_ATBD_NDVI300m-V3).

$$\boldsymbol{NDVI = \frac{NIR-RED}{NIR+RED} \cdot 1.045} \quad \textbf{Eq.4}$$

2.3.2.2 NDVI uncertainty (NDVI_unc)

The uncertainty of the NDVI is calculated by error propagation of the TOC-r uncertainty layers that are delivered with each spectral reflectance. No additional uncertainty is added by calculating the NDVI, since NDVI is calculated using merely arithmetic expressions.

The NDVI uncertainty (ΔNDVI) is calculated from the uncertainties associated with the TOC-r layers (ΔB7, etc.) according to Eq. 5, Eq. 6 and Eq. 7.

meanRED uncertainty

$$\Delta meanRED = \frac{1}{2}\sqrt{\Delta B7^2 + \Delta B8^2} \quad \textbf{Eq. 5}$$

meanNIR uncertainty

$$\Delta meanNIR = \frac{1}{2}\sqrt{\Delta B16^2 + \Delta B18^2} \quad \textbf{Eq. 6}$$

NDVI uncertainty

$$\Delta NDVI = \frac{\sqrt{meanNIR^2 \Delta meanRED^2 + meanRED^2 \Delta meanNIR^2}}{(NIR + RED)^2} \quad \textbf{Eq.7}$$

With ΔX is the uncertainty of X. It is considered that the uncertainties on RED and NIR are not correlated.

For PROBA-V, the uncertainty is derived from the uncertainty of the RED and NIR bands as follows

$$\Delta NDVI = \frac{\sqrt{NIR^2 \Delta RED^2 + RED^2 \Delta NIR^2}}{(NIR + RED)^2}$$ **Eq. 8**

2.3.2.3 Quality flags (QFLAG)

The quality flag of the NDVI 300m V3.0 is a bitwise map that propagates information of each TOC-r input band (see also §3.3.1).

2.3.2.4 Number of observations (NOBS)

Each input TOC-r band has an associated Nobs layer. This indicates the number of clear TOC observations in the dekad time window [t0, t0+10days], where t0 is the date of the dekad provided (see §3.4.3). This information is propagated to the NDVI by taking the minimum value of the available Nobs layers, one per spectral band, for that pixel (4 bands for OLCI, 2 for PROBA-V).

2.4 LIMITATIONS OF THE PRODUCT

All reported limitations of the atmospheric correction and the BRDF correction are also applicable to the NDVI as far as they concern the 4 OLCI or 2 PROBA-V input bands used for the NDVI calculation. These limitations are described in detail in the ATBDs [CGLOPS1_ATBD_S3-AC-V1.1 (only for S3 OLCI) and CGLOPS1_ATBD_BRDFCorrection300m-V1.6 (for both sensors)]. A short summary is given below. Since the PROBA-V data are atmospherically corrected using the same model as for S3 OLCI [PROBA-V C2 ATBD AC], the same limitations are valid.

Limitations related to the atmospheric correction:

- The Simplified Method for Atmospheric Correction (SMAC) (Rahman and Dedieu, 1994) is a parameterization of the Second Simulation of the Satellite Signal in the Solar Spectrum radiative transfer model (6S) (Vermote et al., 1997). For certain conditions, e.g. anisotropic surfaces, the SMAC approximations could lead to an inaccuracy of several percent, especially for high atmospheric turbidity.
- Neither adjacency effects nor terrain slope correction is considered. These effects are, however, secondary at the spatial scale of 300 meters.

Limitations related to the BRDF correction:

- The quality of BRDF descriptor retrievals is fully dependent on the quality of the atmospheric correction and cloud-shadow-snow pixel identification. Although a robust outlier detection is part of the ReBeLS processor, its performance is significantly diminished in periods with a high frequency of cloud occurrence.

- The MCD43P dataset (based on MODIS MCD43A1 and MCD43A2 products) that is used as prior input for the BRDF inversion does not make a distinction for snow and snow-free pixels.
- BRDF descriptors and BRDF adjusted reflectances estimates are provided each dekad even if no observations were available within that dekad (and thus NOBS=0). These estimates are significantly dominated by the MCD43P prior of BRDF descriptors used by ReBeLS. In NDVI V3.0, a quality flag is raised when the MCD43P are gap filled by means of a linear interpolation, and the NDVI is flagged when this is the case for pixels > 55°N and when NOBS=0.

Limitations related to the input data:

- Clear-sky snow observations are considered valid inputs to the BRDF inversion, which results in snow contamination in the NDVI outputs. To mitigate this, in NDVI 300m V3.0 the information on $Nobs_{snow}$ is propagated. Nevertheless, the snow identification in S3 OLCI is known to underperform and clear-sky snow is often misclassified as cloud by the pixel identification scheme (see CGLOPS1_QAR_S3-CloudMask). The snow identification of PROBA-V was much improved in C2, but has still limitations.
- Distinction between land and water is done differently for products based on PROBA-V C2 inputs (static status map bit 3) and on Sentinel-3 (static IdePix land mask). The former only masks open sea/ocean, while the latter masks open sea/ocean, lakes, reservoirs and wide rivers. This leads to an inconsistency on which pixels are retained for TOC-r and NDVI processing for the PROBA-V and Sentinel-3 period, respectively.
- The per-pixel TOC reflectance uncertainties are used in the BRDF inversion. However, for PROBA-V TOC reflectances, these uncertainties are not available, and a 2% uncertainty is assumed for Red and NIR reflectance

2.5 DIFFERENCES WITH PREVIOUS VERSIONS

The differences in algorithms between NDVI 300m V3.0 and the previous versions are summarized in Table 2. Next to the NDVI 300m, a NDVI 1km product also exists. The different versions of the latter are described in Table 3.

Table 2: Algorithm differences between successive versions of the NDVI 300m products

NDVI 300m version	Algorithm summary	Time series available
V1	Derived from PROBA-V C1 S10 products, with incorporation of the Status Map and land mask.	Jan/2014 – Jun/2020 (only PROBA-V)
V2	Angular corrected composite over land pixels using ReBeLS v1.3, derived from Sentinel-3 OLCI TOC data, with a 30-days composite window, updated every 10 days using a sliding window. The normalized TOC V1 reflectances from OLCI bands 7, 8, 9 and 10 are averaged to a meanRED reflectance and the OLCI bands 16, 17 and 18 are averaged to a meanNIR reflectance prior to NDVI computation	From Jul/2020 onwards.
V3	BRDF normalization is done Sentinel-3 TOC V2.3 and PROBA-V C2 S1 TOC data, using ReBeLS v1.6. The number of snow observations in the BRDF compositing window ($Nobs_{snow}$) is propagated to the QFLAG. A quality flag is raised when the BRDF priors are based on interpolated values.	From January/2014 onwards.

Table 3: Algorithm differences between successive versions of the NDVI 1km products

NDVI 1km version	Algorithm summary	Time series available
V3	Derived from SPOT/VGT C3 TOA segments and PROBA-V C1 TOA data; normalized surface reflectance, 10-days composite window (extendable to 16 days), updated every 10 days using a sliding window. A spectral harmonization between VGT1 and VGT2 is performed.	Jan/99 – Jun/2020 (VGT and PROBA-V)

© C-GLOPS Lot1 consortium

3 PRODUCT DESCRIPTION

3.1 FILE NAMING

The **Normalized Difference Vegetation Index (NDVI)** 300m V3.0 product follows the naming standard:

c_gls_<Acronym>_<TIME>_<AREA>_<SENSOR>_V<VERSION>

where

- <Acronym> identifies the product. Here, NDVI300 refers to the Normalized Difference Vegetation Index at 300m resolution
- <TIME> indicates the product reference date. It is expressed in YYYYMMDDhhmm where YYYY, MM, and DD denote the year, the month, and the day, respectively. Three products are delivered per month, i.e. with reference date DD 01, 11 or 21, corresponding to the first day of the dekad. "hhmm" are always "0000".
- <AREA> gives the spatial coverage of the file. For NDVI300, <AREA> is GLOBE, short name for global.
- <SENSOR> gives the name of the sensor used to retrieve the product.
- <VERSION> shows the processing line version used to generate these NDVI products. The version is denoted as M.m.r (e.g. 2.1.1), with 'M' representing the major version (e.g. V2), 'm' the minor version (starting from 0) and 'r' the production run number (starting from 1) (Table 4).

Table 4: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	Recommendations
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated.
Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues, for instance when a reprocessing is necessary due to degraded ancillary data. Later run automatically replaces former	Consider it as a drop-in replacement

© C-GLOPS Lot1 consortium

3.2 FILE FORMAT

The NDVI 300m V3.0 products are distributed as Multi-band Network Common Data Form version 4 (netCDF4) file with metadata attributes compliant with version 1.6 of the Climate & Forecast conventions (CF V1.6) and containing the following layers (netCDF variables):

- NDVI: (angular corrected) normalized difference vegetation index value
- NDVI_unc: uncertainty associated to NDVI
- NOBS: number of TOC reflectances in the dekad time window, minimum over the seven spectral bands used for NDVI calculation (see §2.3.2.1).
- QFLAG: quality flag

3.3 PRODUCT CONTENT

3.3.1 Data File

The physical range and scaling factors of the different NetCDF layers are summarized in Table 5. They are also available in the layer attributes of metadata (Table 9). The physical values (PhyVal) are retrieved from the Digital Numbers (DN) through the following equation:

$$PhyVal = DN * scale_factor + add_offset \quad \textbf{Eq 9}$$

Note that the **NDVI** values are restricted to the range [-0.08; 0.92]. The upper digital values (DN values above 250) are available for "flags", i.e. dedicated values to indicate special situations (see Table 6). The NDVI product is distributed in byte, as we consider the uncertainties caused by radiometric calibration, atmospheric correction and BRDF correction to be larger than the precision of 0.004 (the radiometric resolution in the current product).

Table 5: Minimum, maximum values and associated resolution for NDVI product layers

Variable	Physical Minimum	Physical Maximum	Max DN value	scale_factor	add_offset	NoData value(s)
NDVI	-0.08	0.92	250	0.004	-0.08	252, 253, 254, 255
NDVI_unc	0	1.0	1000	0.001	0	65535
NOBS	0	32	32	1	0	255
QFLAG	N/A	N/A	254	1	0	255

© C-GLOPS Lot1 consortium

Table 6: Description of the quality flags embedded into the NDVI layer

Flag value	Flag name	Description
252	Unknown	NOBS = 0 and the BRDF priors are interpolated (QFLAG = 128, see Table 7). This flag value only occurs above 55°N.
253	Snow	NOBS > 0 and NOBS_SNOW ≥ ½ NOBS
254	Water	Water
255	Missing	Where TOC-r is out of range (QFLAG = 64, see Table 7) or where no TOC-r for at least one of the red or NIR bands is available

The **NDVI_unc** variable contains the uncertainty associated to the NDVI estimation. It is calculated from the uncertainties of the normalized reflectances (Eq.5, Eq.6, Eq.7 and Eq.8) which are propagated from the uncertainties associated to the BRDF model parameters found via the inversion. NDVI_unc is replaced by a flag value for water pixels (if NDVI = 254, NDVI_unc = -2) and for other pixels where the NDVI is flagged (if NDVI = 252 or NDVI = 253 or NDVI = 255, NDVI_unc = -1).

The **NOBS** variable contains the minimum number of clear TOC observations in the dekad time window, i.e. [t0, t0+10days], where t0 is the date of the dekad provided (see §3.4.3), over the spectral band used for the NDVI calculation. When NOBS is equal to zero, QFLAG bit 0 is set to 1.

The **QFLAG** (quality flag) of the NDVI 300m V3.0 describes the BRDF inversion quality in the red and NIR bands for all land pixels. The QFLAG layer is bit-wise coded according to the information summarized in Table 7.

The different bits are described in detail below:

- Bit 0 = 1 (QFLAG = 1): if, at least, for one of the input bands for NDVI calculation, no observations were found within the compositing window. This means that the NDVI is based on a TOC-r value based on prior information only for at least one of the input bands. The NDVI should be treated with caution.
- Bit 1 = 1 (QFLAG = 2): at least one of the observations in the compositing window for any of the TOC-r input bands is classified as ‘snow’. If half, or more than half, of the observations in the compositing window are classified as ‘snow’, the NDVI and NDVI_unc are not calculated, and a flag value (DN=253) is added in these layers.
- Bit 2 = 1 (QFLAG = 4): at least one of the red TOC-r input bands has a quality ‘warning’ [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. It means that the goodness of fit estimation between observations and model is adequate, but results should be taken with caution.
- Bit 3 = 1 (QFLAG = 8): at least one of the red TOC-r input bands has a quality ‘extreme warning’ [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. It means that the goodness of fit estimation between observations and model is not adequate.

© C-GLOPS Lot1 consortium

- Bit 4 = 1 (QFLAG = 16): at least one of the NIR TOC-r input bands has a quality 'warning' [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. It means that the goodness of fit estimation between observations and model is adequate, but results should be taken with caution.
- Bit 5 = 1 (QFLAG = 32): at least one of the NIR TOC-r input bands has a quality 'extreme warning' [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]. It means that the goodness of fit estimation between observations and model is not adequate.
- Bit 6 = 1 (QFLAG = 64): at least one the input TOC-r values was out of range (TOC-r < 0 or TOC-r >1). Also, here, the NDVI and NDVI_unc are not calculated (DN=255), and a flag value is added in these layers (NDVI = 255, NDVI_unc=-1).
- Bit 7 = 1 (QFLAG = 128): the MCD43P climatology (BRDF priors) are gap filled by means of a linear interpolation (see CGLOPS1_ATBD_BRDFCorrection300m-V1.6). If NOBS = 0 and the pixel is located > 55°N, the NDVI and NDVI_unc are not calculated, and a flag value (DN=252 for NDVI, -1 for NDVI_unc) is added in these layers.

Table 7: Bit number, description and value of the quality flags.

Bit	Description	Value
0	No observations found in the compositing window in, at least, one of the red or NIR bands	1
1	At least one of the observations in the compositing window is flagged as 'snow'	2
2	At least one TOC-r input band for red has quality 'warning'	4
3	At least one TOC-r input band for red has quality 'extreme warning'	8
4	At least one TOC-r input band for NIR has quality 'warning'	16
5	At least one TOC-r input band for NIR has quality 'extreme warning'	32
6	TOC-r out of range (TOC-r < 0 | TOC-r > 1) for at least one band	64
7	The BRDF MCD43P priors are gap filled	128

3.3.2 Metadata

The products include metadata on 2 levels: file level and layer level. The metadata in Table 8 is included on file level while the attributes in Table 9 are provided at layer level.

Table 8: Description of file attributes for an OLCI-based NDVI file

Attribute name	Description	Data Type	Example(s)
Conventions	Version of the CF-Conventions used	String	CF-1.6
Title	A description of the contents of the file	String	10-daily Normalized Difference Vegetation Index 300M: GLOBE 2024-06-01-01T00:00:00Z
Institution	The name of the institution that produced the product	String	VITO NV
Source	The method of production of the original data	String	Derived from EO satellite imagery
History	A global attribute for an audit trail. One line, including date in ISO-8601 format, for each invocation of a program that has modified the dataset.	String	2024-06-12 Processing line BRDF v1.6 2024-06-12 Processing line NDVI v3.0
References	Published or web-based references that describe the data or methods used to produce it.	String	https://land.copernicus.eu/en/products/vegetation
archive_facility	Specifies the name of the institution that archives the product	String	VITO
product_version	Version of the product (VM.m.r)	String	V3.0.1
time_coverage_start	Start date and time of the total coverage of the data for the product.	String	2024-06-01T00:00:00Z
time_coverage_end	End date and time of the total coverage of the data for the product.	String	2024-06-10T23:59:59Z
platform	Name(s) of the orbiting platform(s)	String	Sentinel3A,Sentinel3B
sensor	Name(s) of the sensor(s) used	String	OLCI
identifier	Unique identifier for the product	String	urn:cgls:global:ndvi300_v3_333m:NDVI_202406010000_GLOBE_OLCI_V3.0.1
parent_identifier	Identifier of the product collection (time series) for the product in Copernicus Global Land Service metadata catalogue.	String	urn:cgls:global:ndvi300_v3_333m

© C-GLOPS Lot1 consortium

Attribute name	Description	Data Type	Example(s)
long_name	Extended product name	String	Normalized Difference Vegetation Index
orbit_type	Orbit type of the orbiting platform(s)	String	LEO
processing_level	Product processing level	String	L3
processing_mode	Processing mode used when generating the product (Near-Real Time, Consolidated or Reprocessing)	String	Near Real Time
copyright	Text to be used by users when referring to the data source of this product in publications (copyright notice)	String	Copernicus Service information 2025

© C-GLOPS Lot1 consortium

Table 9: Description of layer attributes

Attribute	Description	Data Type	Examples for NDVI layer
CLASS	Dataset type		
long_name	A descriptive name that indicates a variable's content. This name is not standardized. Required when a standard name is not available.	String	Normalized Difference Vegetation Index
scale_factor	Multiplication factor for the variable's contents that must be applied in order to obtain the real values. Omit in case the scale is 1.	Float	0.004
add_offset	Number to be added to the variable's contents (after applying scale_factor) that must be applied in order to obtain the real values. Omit for offset 0.	Float	-0.08
units	Units of the variable's content, taken from Unidata's udunits library. Empty or omitted for dimensionless variables. Fractions should be indicated by scale_factor.	String	
valid_range	Smallest and largest values for the variable. Missing data is to be represented by one or several values outside of this range.	string	0, 250
_FillValue	Single value used to represent missing or undefined data and to pre-fill memory space in case a non-written part of data is read back. Value must be outside of valid_range.	string	255
missing_value	Single value used to represent missing or undefined data, for applications following older versions of the standards. Value must be outside of valid_range.	string	255
grid_mapping	Reference to the grid mapping variable	String	crs
flag_values	Provides a list of the flag values. Used in conjunction with flag_meanings.	string	252, 253, 254, 255
flag_meanings	Descriptive words or phrases for each flag value.	String (blank separated list)	unknown, snow, water, missing

© C-GLOPS Lot1 consortium

3.4 PRODUCT CHARACTERISTICS

3.4.1 Projection and Grid Information

The product is displayed in a regular latitude/longitude grid (plate carrée) with the ellipsoid WGS 1984 (Terrestrial radius=6378km). The resolution of the grid is 1/336°.

The reference is the centre of the pixel. It means that the longitude of the upper left corner of the pixel is (pixel_longitude – angular_resolution/2.)

3.4.2 Spatial Information

The NDVI products are provided from longitude 180°W to 180°E and latitude 85°N to 65°S. For areas that are unavailable in the input data, the NDVI pixel value is set to Missing (255).

Note that the PROBA-V input data only covers the area from latitude 75°N to 65°S. Hence, this is the true coverage of the NDVI for the period 2014 – 2018.

3.4.3 Temporal Information

Three NDVI products are produced per month. The temporal information "YYYYMMDDhhmm" in the filename corresponds to day 1, day 11 and day 21 of each month and are relevant for the period covering up to the 10th, 20th and end of the month respectively.

For example, the file c_gls_NDVI300_202007010000_GLOBE_OLCI_V3.0.1.nc corresponds to the accumulation period from 11/06/2020 to 10/07/2020.

The BRDF-correction uses additional data for outlier detection. For the NRT processing, a total period of 30 days is used, and for the archive processing, an entire year was used.

3.5 DATA POLICIES

The Copernicus component is governed by Regulation (EU) No 2021/696 of the European Parliament and of the Council of 28 April 2021 establishing the Union Space Programme and the European Union Agency for the Space Programme.

The Copernicus land monitoring products and services are made available on a principle of **full, open and free access**, as established by the Commission Delegated Regulation (EU) No 1159/2013 of 12 July 2013.

How to cite and reference the products and services of the Copernicus Land Monitoring Service is explained here: https://land.copernicus.eu/en/data-policy.

3.6 ACCESS AND CONTACTS

The NDVI 300m V3.0 products are available through the Copernicus Land Monitoring Service website at the address: https://land.copernicus.eu/en/products/vegetation?tab=vegetation_indices with scientific and technical contact information (helpdesk) on https://land.copernicus.eu/en/contact-service-helpdesk.

© C-GLOPS Lot1 consortium

4 QUALITY ASSESSMENT

The Quality Assessment Report [CGLOPS1_QAR_NDVI300m-V3] contains a detailed analysis of the quality of the NDVI 300m V3 product. The quality assessment (QA) focused on (1) the occurrences of quality flags and uncertainties, (2) product completeness, (3) visual comparison with reference datasets, (4) statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI, (5) statistical and spatial consistency with NDVI 300m V1 and NDVI 300m V2, (6) statistical and spatial consistency with NDVI 1km V3, (7) statistical and spatial consistency with MCD43A4 V6.1, and (8) temporal consistency. It is based on the full back-processed time series 2014 – 2024 for most of the analyses but used the year 2019 for statistical and spatial comparison with reference datasets NDVI 300m V1, NDVI 1km V3, and MCD43A4 V6.1, and the year 2023 for statistical and spatial comparison with NDVI 300m V3.

For the intercomparison between PROBA-V and Sentinel-3/OLCI, a test dataset of NDVI 300m V3 PROBA-V data for the year 2019 was used. This dataset is not available in the disseminated product, as the switch between PROBA-V and Sentinel-3/OLCI is made at the start of 2019.

The results of the QAR are summarised here:

1. The NOBS=0 quality flag and high NDVI uncertainties occur mainly during the Northern hemisphere wintertime, at high latitudes, and in areas with persistent cloud cover (e.g. the wet tropics). There is no clear spatial or temporal pattern for the 'Warning' and 'Extreme warning' occurrences. There are more NOBS=0 and less 'Warning', 'Extreme warning', and snow flag occurrences for Sentinel-3/OLCI compared to PROBA-V, related to higher NDVI uncertainties in Sentinel-3/OLCI and differences in cloud/snow masking algorithm. The NDVI uncertainty decreases exponentially with increasing NOBS.

2. The product completeness of the NDVI 300m V3 back-processed dataset is very high, related to the long accumulation period and the use of a climatology in the BRDF inversion. In NRT processing, the spatial and temporal completeness are more comparable to the product completeness of NDVI 300m V1, NDVI 300m V2, NDVI 1km V3, and MCD43A4 V6.1, with more gaps at high latitudes during the winter period and in areas with persistent cloud cover.

3. NDVI 300m V3 visual quicklooks are similar to the reference products with respect to sharpness and spatial patterns. Differences in completeness between NDVI 300m V3 and NDVI 300m V2 can be explained by the inclusion of a snow and interpolated prior quality flag in V3, and the longer accumulation window.

4. There is good statistical and spatial consistency between PROBA-V and Sentinel-3/OLCI for NDVI 300m V3. Some present differences are largest for forest biome types, related to

differences in cloud/snow detection and the impact of the MODIS prior when no valid input TOC data are available. The NDVI uncertainties of Sentinel-3/OLCI are on average a factor two larger than those of PROBA-V.

5. Overall, there is good consistency between NDVI 300m V3 and the previous CLMS NDVI 300m versions. There is a small negative bias with NDVI 300m V1, strongly linked to vegetation densities and caused by anisotropy effects and differences in atmospheric correction. The consistency with NDVI 300m V2 is much better, and small differences are explained by differences in the ReBeLS BRDF-correction algorithm.

6. Large statistical and spatial consistency is found between NDVI 300m V3 and NDVI 1km V3. Differences are mostly unsystematic. Therefore, the NDVI 1km V3 dataset can be used to extend the NDVI 300m V3 to the past (i.e. 1999).

7. There is a small negative bias between NDVI 300m V3 and MCD43A4 V6.1, most likely related to differences in sensor and processing techniques. The effects are largest for forest biomes and at high latitudes.

8. NDVI 300m V3 shows similar temporal behaviour compared to the reference datasets. The temporal profiles are more stable than for NDVI 300m V1 and show higher completeness in winter due to the larger accumulation window in the BRDF normalisation. The temporal smoothness is large and the Hovmöller plots show that the temporal consistency between PROBA-V and Sentinel-3/OLCI is improved in NDVI 300m V3 compared to the combination of NDVI 300m V1 and NDVI 300m V2. The stability of the NDVI 300m V3 product is within the requirements (see §7.1).

In conclusion, the NDVI 300m V3 product shows high product completeness, good stability, and large statistical, spatial, and temporal consistency with the reference datasets. Some differences are expected because of differences in sensor, processing approaches (especially BRDF normalisation), and flagging of bad data. Most discrepancies are minimal and unsystematic. Since the PROBA-V data from 2014 – 2018 and the Sentinel-3/OLCI data from 2019 – onwards are now both corrected for BRDF effects, NDVI differences between the two sensors are minimal.

5 PRODUCT USAGE

5.1 RECOMMENDATIONS ON USING THE QUALITY FLAG

Based on the extensive quality assessment [CGLOPS1_QAR_NDVI300m-V3], users are recommended to exclude NDVI values with 'Extreme warning', since the estimation of goodness of fit indicates that the agreement between observations and the BRDF model fit cannot be considered adequate (see [CGLOPS1_ATBD_BRDFCorrection300m-V1.6]).

In addition, in order to exclude NDVI values with large uncertainty, the user can decide (i) to filter on a fixed NDVI_unc value, or (ii) (recommended) to exclude NDVI values where NOBS=0 (or QFLAG bit 0 raised).

5.2 SUITABILITY FOR SPECIFIC SECTORS AND APPLICATIONS

NDVI can be used as a proxy for vegetation amount, health and density and can therefore be used in a wide range of environmental applications such as e.g. agriculture, forestry, ecology, climate and environmental management.

Because the NDVI is an arithmetic index based on spectral bands, it is dependent on the spectral definitions of a sensor. Therefore, the temporal evolution of NDVI and its change over time at intra-annual and interannual time scales is usually of interest.

Huang et al. (2021) gives an overview of the fields in which NDVI is most popularly used and also discusses its main limitations. The fields relate mainly to vegetation and crop monitoring (e.g. biomass, vegetation cover, vegetation dynamics, vegetation growth, net primary production, phenology, photosynthesis, fire), but also climate and energy-related applications (temperature (for the estimation of the land surface emissivity), evapotranspiration, drought, precipitation, transpiration). Many applications focus on time series, trends and variability of NDVI.

NDVI is also widely used for agricultural crop monitoring (e.g. Baruth et al., 2008; Becker-Reshef et al., 2018; Fritz et al., 2019), yield predictions (e.g. Fuentes et al., 2025; Gao et al., 2018; D. M. Johnson et al., 2021; H. E. Johnson et al., 2018; Shammi & Meng, 2023) and crop identification and mapping (e.g. de la Casa et al., 2018; Delrue et al., 2013; Tian et al., 2019; Wardlow & Egbert, 2008, 2010).

Medium resolution NDVI from various sources underpin operational crop dashboards, enabling seasonal anomaly tracking, spatial masks for crop areas, and admin unit timeseries for early warning. Examples are JRC MARC bulletins (https://joint-research-centre.ec.europa.eu/monitoring-agricultural-resources-mars/jrc-mars-bulletin_en), USDA/NASA GLAM & Crop Explorer (https://glam1.gsfc.nasa.gov/,

https://ipad.fas.usda.gov/cropexplorer/imageview.aspx?regionid=world&product=modis_ndvi_8day), GEOGLAM Crop Monitor (https://www.cropmonitor.org/).

FAO ASIS provides an NDVI-based service on early warning of agricultural drought (https://asis.apps.fao.org/).

A drought assessment using NDVI overview can be found in (Karnieli et al., 2010). CLMS NDVI was also used for agricultural drought assessment by (De Vos et al., 2025).

An overview of the use of NDVI for land degradation assessment is provided by (Yengoh et al., 2014), who identified land-use and land-cover change, drought and drought early warning, desertification, soil erosion, soil salinization, vegetation burning, soil organic carbon, biodiversity monitoring and conservation and monitoring ecosystem resilience as fields in which NDVI can be used.

NDVI was reported to be useful in animal ecology for the prediction of herbivore and non-herbivore distribution, abundance and life history traits in space and time (Pettorelli et al., 2011).

© C-GLOPS Lot1 consortium

6 REFERENCES

Baret, F., Guyot, G., 1991. Potentials and limits of vegetation indices for LAI and APAR assessment. Remote Sens. Environ. 35, 161–173. https://doi.org/10.1016/0034-4257(91)90009-U

Baruth, B., Royer, a, Klisch, A., Genovese, G., 2008. The use of remote sensing within the MARS crop yield monitoring system of the European Commission. Proceedings ISPRS 27, 935–940.

Becker-Reshef, I., Justice, C., Whitcraft, A.K., Jarvis, I., 2018. Geoglam: A Geo Initiative on Global Agricultural Monitoring, in: IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium. IEEE, pp. 8155–8157. https://doi.org/10.1109/IGARSS.2018.8517575

Carreiras, J.M.B., Pereira, J.M.C., Shimabukuro, Y.E., Stroppiana, D., 2003. Evaluation of compositing algorithms over the Brazilian Amazon using SPOT-4 VEGETATION data. Int. J. Remote Sens. 24, 3427–3440. https://doi.org/10.1080/0143116021000021251

de la Casa, A., Ovando, G., Bressanini, L., Martínez, J., Díaz, G., Miranda, C., 2018. Soybean crop coverage estimation from NDVI images with different spatial resolution to evaluate yield variability in a plot. ISPRS Journal of Photogrammetry and Remote Sensing 146, 531–547. https://doi.org/10.1016/j.isprsjprs.2018.10.018

De Vos, K., Gebruers, S., Degerickx, J., Iordache, M.-D., Keune, J., Di Giuseppe, F., Vilela Pereira, F., Wouters, H., Swinnen, E., Van Rossum, K., Tits, L., 2025. Predicting below-average NDVI anomalies for agricultural drought impact forecasting. Remote Sens Environ 330, 114980. https://doi.org/10.1016/j.rse.2025.114980

Delrue, J., Bydekerke, L., Eerens, H., Piccard, I., Swinnen, E., Gilliams, S., 2013. Crop mapping in countries with small-scale farming: a case study for West Shewa, Ethiopia. Int J Remote Sens 34, 2566–2582.

Duggin, M., Piwinski, D., Whitehead, V., Ryland, G., 1982. Evaluation of NOAA-AVHRR data for crop assessment. Appl. Opt. 21, 1873–1875.

Eklundh, L., 1995. Noise estimation in NOAA AVHRR maximum-value composite NDVI images. Int. J. Remote Sens. 16, 2955–2962.

Fritz, S., See, L., Bayas, J.C.L., Waldner, F., Jacques, D., Becker-Reshef, I., Whitcraft, A., Baruth, B., Bonifacio, R., Crutchfield, J., Rembold, F., Rojas, O., Schucknecht, A., Van der Velde, M., Verdin, J., Wu, B., Yan, N., You, L., Gilliams, S., Mücher, S., Tetrault, R., Moorthy, I., McCallum, I., 2019. A comparison of global agricultural monitoring systems and current gaps. Agric Syst 168, 258–272. https://doi.org/10.1016/j.agsy.2018.05.010

Fuentes, I., Al-Shammari, D., Al-Nasrawi, A.K.M., Wang, Y., Wang, J., Lebrini, Y., Chen, Y., Jones, B.G., Bishop, T.F.A., 2025. The normalised difference vegetation index as an analytic tool for wheat crop yield prediction: A review and meta-analysis. Precis Agric 26, 55. https://doi.org/10.1007/s11119-025-10247-z

Gao, F., Anderson, M., Daughtry, C., Johnson, D., 2018. Assessing the Variability of Corn and Soybean Yields in Central Iowa Using High Spatiotemporal Resolution Multi-Satellite Imagery. Remote Sens (Basel) 10, 1489. https://doi.org/10.3390/rs10091489

Gutman, G.G., 1989. On the relationship between monthly mean and maximum-value composite normalized vegetation indices. Int. J. Remote Sens. 10, 1317–1325.

Holben, B.N., 1986. Characteristics of maximum-value composite images from temporal AVHRR data. Int. J. Remote Sens. 7, 1417–1434.

Huang, S., Tang, L., Hupy, J.P., Wang, Y., Shao, G., 2021. A commentary review on the use of normalized difference vegetation index (NDVI) in the era of popular remote sensing. J For Res (Harbin). https://doi.org/10.1007/s11676-020-01155-1

Johnson, D.M., Rosales, A., Mueller, R., Reynolds, C., Frantz, R., Anyamba, A., Pak, E., Tucker, C., 2021. USA Crop Yield Estimation with MODIS NDVI: Are Remotely Sensed Models Better than Simple Trend Analyses? Remote Sens (Basel) 13, 4227. https://doi.org/10.3390/rs13214227

Johnson, H.E., Gustine, D.D., Golden, T.S., Adams, L.G., Parrett, L.S., Lenart, E.A., Barboza, P.S., 2018. NDVI exhibits mixed success in predicting spatiotemporal variation in caribou summer forage quality and quantity. Ecosphere 9. https://doi.org/10.1002/ecs2.2461

Karnieli, A., Agam, N., Pinker, R.T., Anderson, M., Imhoff, M.L., Gutman, G.G., Panov, N., Goldberg, A., 2010. Use of NDVI and land surface temperature for drought assessment: Merits and limitations. J Clim 23. https://doi.org/10.1175/2009JCLI2900.1

Meyer, D., Verstraete, M., Pinty, B., 1995. The effect of surface anisotropy and viewing geometry on the estimation of NDVI from AVHRR. Remote Sens. Rev. 12, 3–27.

Moody, A., Strahler, A., 1994. Characteristics of composited AVHRR data and problems in their classification. Int. J. Remote Sens. 15, 3473–3491.

Pettorelli, N., Ryan, S., Mueller, T., Bunnefeld, N., Jedrzejewska, B., Lima, M., Kausrud, K., 2011. The Normalized Difference Vegetation Index (NDVI): unforeseen successes in animal ecology. Clim Res 46, 15–27. https://doi.org/10.3354/cr00936

Rahman, H., Dedieu, G., 1994. SMAC: a simplified method for the atmospheric correction of satellite measurements in the solar spectrum. Int. J. Remote Sens. 15, 123–143. https://doi.org/10.1080/01431169408954055

Rouse, J., Haas, R., Schell, J., Deering, D., Harlan, J., 1974. Monitoring the vernal advancement of retrogradation of natural vegetation. NASA/GSFC.

Shammi, S.A., Meng, Q., 2023. Modeling crop yield using NDVI-derived VGM metrics across different climatic regions in the USA. Int J Biometeorol 67, 1051–1062. https://doi.org/10.1007/s00484-023-02478-4

Tian, H., Huang, N., Niu, Z., Qin, Y., Pei, J., Wang, J., 2019. Mapping winter crops in China with multi-source satellite imagery and phenology-based algorithm. Remote Sens (Basel) 11. https://doi.org/10.3390/rs11070820

Tucker, C., 1979. Red and photographic infrared linear combinations for monitoring vegetation. Remote Sens. Environ. 150, 127–150.

Tucker, C.J., Holben, B.N., Elgin, J.H., McMurtrey, J.E., 1981. Remote sensing of total dry-matter accumulation in winter wheat. Remote Sens. Environ. 11, 171–189. https://doi.org/10.1016/0034-4257(81)90018-3

Vermote, E.F., Tanré, D., Deuzé, J.L., Herman, M., Morcrette, J.J., 1997. Second simulation of the satellite signal in the solar spectrum, 6s: an overview. IEEE Trans. Geosci. Remote Sens. 35,

675–686. https://doi.org/10.1109/36.581987

Wardlow, B.D., Egbert, S.L., 2008. Large-area crop mapping using time-series MODIS 250 m NDVI data: An assessment for the U.S. Central Great Plains. Remote Sens Environ 112, 1096–1116. https://doi.org/10.1016/j.rse.2007.07.019

Yengoh, G.T., Dent, D., Olsson, L., Tengberg Compton, A.E., Tucker, J., 2014. The use of the Normalized Difference Vegetation Index (NDVI) to assess land degradation at multiple scales: a review of the current status, future trends, and practical considerations.

7 ANNEX: REQUIREMENTS

According to the applicable documents [AD1], the requirements relevant for CLMS global NDVI product are described below.

7.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300m,
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	regular latitude/longitude grid
Geographic Coverage	Global
Temporal resolution	10-day period (dekad: days 1-10, 11-20, 21 end of month)
Timeliness	Within 2 days (optimally 1 day) after the end of each dekad

NDVI – NORMALIZED DIFFERENCE VEGETATION INDEX	
The Normalized Difference Vegetation Index (NDVI) is a dimensionless parameter that is chlorophyll sensitive and indicative for the greenness of the biomes and is expressed as the difference between the near-infrared and red bands, normalized to the summation.	
Uncertainty (2-sigma)	Threshold: 5% Goal: 2.5%
Stability (per decade, i.e. 10 years)	Threshold: 3% Goal: 1.5%

7.2 FURTHER REQUIREMENTS

7.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

© C-GLOPS Lot1 consortium

7.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.

Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

7.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NetCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the Contracting Authority during the Framework Contract.

7.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

7.2.5 Input data

Copernicus Sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, PROBA-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

7.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic